

Week-7: To Do List

1. Using Scheme diagram, Create tables by properly specifying the primary keys and the foreign keys.
2. Insert appropriate records in each table.
3. Find the pnames of parts for which there is some supplier.
4. Find the snames of suppliers who supply every part.
5. Find the snames of suppliers who supply every red part.
6. Find the pnames of parts supplied by Acme Widget Suppliers and by no one else.
7. Find the sids of suppliers who charge more for some part than the average cost of that part (averaged over all the suppliers who supply that part).
8. For each part, find the sname of the supplier who charges the most for that part.

ANSWERS: -

```
1 • create database sup;
2 • use sup;
3
4 • CREATE TABLE Supplier (
5     sid INT PRIMARY KEY,
6     sname VARCHAR(50),
7     address VARCHAR(100)
8 );
9
10 • CREATE TABLE Parts (
11     pid INT PRIMARY KEY,
12     pname VARCHAR(50),
13     color VARCHAR(20)
14 );
15
16 • CREATE TABLE Catalog (
17     sid INT,
18     pid INT,
19     cost INT,
20     PRIMARY KEY (sid, pid),
21     FOREIGN KEY (sid) REFERENCES Supplier(sid),
22     FOREIGN KEY (pid) REFERENCES Parts(pid)
23 );
24
```

```

25 • INSERT INTO Supplier VALUES
26     (10001, 'Acme Widget', 'Bangalore'),
27     (10002, 'Johns', 'Mysore'),
28     (10003, 'Reliance', 'Chennai'),
29     (10004, 'Tata', 'Mumbai');
30
31 • INSERT INTO Parts VALUES
32     (20001, 'Book', 'Red'),
33     (20002, 'Charger', 'Green'),
34     (20003, 'Mobile', 'Black'),
35     (20004, 'Pen', 'Red'),
36     (20005, 'Pencil', 'Blue');
37
38 • INSERT INTO Catalog VALUES
39     (10001, 20001, 50),
40     (10001, 20002, 60),
41     (10001, 20003, 150),
42     (10001, 20004, 35),
43     (10002, 20001, 55),
44     (10002, 20004, 45),
45     (10003, 20003, 120),
46     (10003, 20005, 40),
47     (10004, 20001, 65);
48
49 • SELECT DISTINCT p.pname
50     FROM Parts p
51     JOIN Catalog c ON p.pid = c.pid;
52

```

	pname
►	Book
	Charger
	Mobile
	Pen
	Pencil

```

53 • SELECT s.sname
54 FROM Supplier s
55 WHERE NOT EXISTS (
56     SELECT *
57     FROM Parts p
58     WHERE NOT EXISTS (
59         SELECT *
60         FROM Catalog c
61         WHERE c.sid = s.sid AND c.pid = p.pid
62     )
63 );

```

Result Grid
sname

```

65 • SELECT s.sname
66 FROM Supplier s
67 WHERE NOT EXISTS (
68     SELECT *
69     FROM Parts p
70     WHERE p.color = 'Red'
71     AND NOT EXISTS (
72         SELECT *
73         FROM Catalog c
74         WHERE c.sid = s.sid AND c.pid = p.pid
75     )
76 );
77
78 • SELECT p.pname

```

Result Grid
sname
Acme Widget
Johns

```

78 • SELECT p.pname
79 FROM Parts p
80 WHERE p.pid IN (
81     SELECT c.pid
82     FROM Catalog c
83     JOIN Supplier s ON c.sid = s.sid
84     WHERE s.sname = 'Acme Widget'
85 )
86 AND p.pid NOT IN (
87     SELECT c.pid
88     FROM Catalog c
89     JOIN Supplier s ON c.sid = s.sid
90     WHERE s.sname <> 'Acme Widget'
91 );
92

```

Result Grid		Filter Rows:	Export:
	pname		
►	Charger		

```

93 • SELECT DISTINCT c.sid
94 FROM Catalog c
95 WHERE c.cost >
96     (SELECT AVG(c2.cost)
97      FROM Catalog c2
98      WHERE c2.pid = c.pid);
99

```

Result Grid		Filter Rows:
	sid	
►	10001	
	10002	
	10004	

```

100 • SELECT p.pid, s.sname
101 FROM Parts p
102 JOIN Catalog c ON p.pid = c.pid
103 JOIN Supplier s ON c.sid = s.sid
104 WHERE c.cost =
105     (SELECT MAX(c2.cost)
106      FROM Catalog c2
107      WHERE c2.pid = p.pid);
108

```

Result Grid			Filter Rows:
	pid	sname	
▶	20002	Acme Widget	
	20003	Acme Widget	
	20004	Johns	
	20005	Reliance	
	20001	Tata	